

calc

Sheet: Inputs

Parameter	Value	Unit	Source
Positive electrode thickness	75.6	um	parameter set
Negative electrode thickness	108	um	design override
Separator thickness	7	um	design override
Positive current collector thickness	8	um	design override
Negative current collector thickness	6	um	design override
Positive porosity	0.335	-	parameter set
Negative porosity	0.25	-	parameter set
Positive density (LNMO)	4400	kg/m3	LNMO.json
Negative density (graphite)	1657	kg/m3	parameter set
Electrode area	0.1027	m2	calc-energy
Nominal capacity	4.5	Ah	LNMO.json

Sheet: Capacity_Energy

Quantity	Value	Unit	Formula/Source
1C discharge capacity	6.37219557690128	Ah	run-pyamm 1C_discharge
Discharge energy	26.47885830600313	Wh	integral V*I dt
Cell mass (contract)	0.04439263676900001	kg	sum layer (1-porosity)*density*area
Cell volume	2.101242e-05	m3	sum layer thickness*area
Midpoint voltage	4.131295823917646	V	calc-energy

Sheet: Energy_Density

Quantity	Value	Unit	Formula/Source
Gravimetric energy density	596.4696002129273	Wh/kg	energy/mass
Volumetric energy density	1260.152724246095	Wh/L	energy/volume (electrolyte excluded)

Sheet: NP_Mass

Layer	Areal mass (kg/m2)	Mass (g)	Source
Positive electrode	0.2212056	22.71781512	calc-energy layer_kg_m2
Negative electrode	0.134217	13.7840859	calc-energy layer_kg_m2
Positive CC	0.0216	2.21832	calc-energy layer_kg_m2
Negative CC	0.05376	5.521152000000001	calc-energy layer_kg_m2
Separator	0.00147287	0.151263749	calc-energy layer_kg_m2
N/P ratio (negative/positive areal capacity)	1.037492475206717		0.895*c_max_neg*L*eps_am*F/3600 / (cap/area)

Sheet: Process

Parameter	Value	Unit	Formula/Source
Positive areal density	221.2056	g/m2	$t \cdot (1 - \text{por}) \cdot \text{density}$
Negative areal density	134.217	g/m2	$t \cdot (1 - \text{por}) \cdot \text{density}$
Positive compaction density	2.926	g/cm3	$\text{density} \cdot (1 - \text{por}) / 1000$
Negative compaction density	1.24275	g/cm3	$\text{density} \cdot (1 - \text{por}) / 1000$
Electrolyte fill amount	6.85411584	g	pore volume x 1.2 g/cm3 (literature)
Formation recommendation	0.1C CC to 4.7V, 25C, 2 cycles	-	design-recommended (annotated)